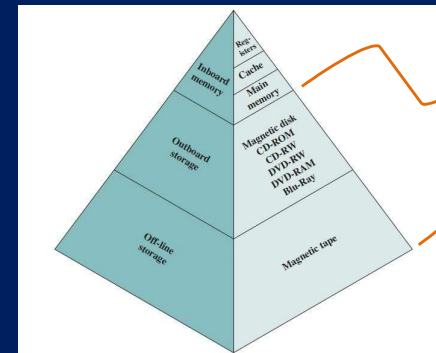


External Memory and Semiconductor Type Storages

pmarzan

Memory Hierarchy



External
Memory
Storages

- ✓ SSD
- ✓ Flash Drive
- ✓ SD Card

Computer Storages

- Electronic Storages
 - Magnetic Storages
 - Optical storages
- Semiconductor Type Storages
 - Volatile Storages
 - Non-Volatile Storages

Magnetic Storage

- Stores data in magnetic form
- Affected by magnetic fields
- Has high storage capacity
- Doesn't use a laser to read/write data

Examples: Hard disk , Floppy disk, Magnetic tape etc.

Magnetic Tapes



Magnetic Disk (Floppy Diskettes)



Magnetic Disk (Zip Diskettes)



Magnetic Disk (Hard Disk)



Optical Storage

- Stores data optically, uses laser to read/write
- Not affected by magnetic fields
- Has less storage than a hard disk
- Data accessing is high, compared to a floppy disc

Examples: CD-ROM, CD-R, CD-RW, DVD etc.

Optical Disk

CD

Compact Disk. A nonerasable disk that stores digitized audio information. The standard system uses 12-cm disks and can record more than 60 minutes of uninterrupted playing time.

CD-ROM

Compact Disk Read-Only Memory. A nonerasable disk used for storing computer data. The standard system uses 12-cm disks and can hold more than 650 Mbytes.

CD-R

CD Recordable. Similar to a CD-ROM. The user can write to the disk only once.

CD-RW

CD Rewritable. Similar to a CD-ROM. The user can erase and rewrite to the disk multiple times.

Optical Disk

DVD

Digital Versatile Disk. A technology for producing digitized, compressed representation of video information, as well as large volumes of other digital data. Both 8 and 12 cm diameters are used, with a double-sided capacity of up to 17 Gbytes. The basic DVD is read-only (DVD-ROM).

DVD-R

DVD Recordable. Similar to a DVD-ROM. The user can write to the disk only once. Only one-sided disks can be used.

DVD-RW

DVD Rewritable. Similar to a DVD-ROM. The user can erase and rewrite to the disk multiple times. Only one-sided disks can be used.

Blu-ray DVD

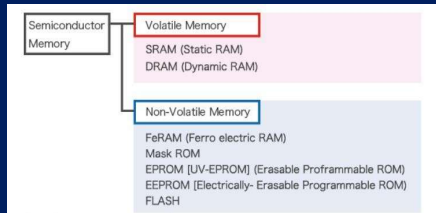
High-definition video disk. Provides considerably greater data storage density than DVD, using a 405-nm (blue-violet) laser. A single layer on a single side can store 25 Gbytes.

Other External Storages

- **SSD (Solid State Drive)**
- **Flash Drive** (USB flash drive -- also known as a USB stick, USB thumb drive, or pen drive)
- **SD Card** (Secure Digital Card)



Semiconductor Type Storage



- **Volatile Memory** – this is a memory storage that loses its data once power is cut off
- **Non-volatile Memory** – this is a memory that retains data even without power

RAM (Random Access Memory) : Enables Read/Write of stored contents
 ROM (Read Only Memory) : Allows only Read operation

Technical Indicator of Memory Storages

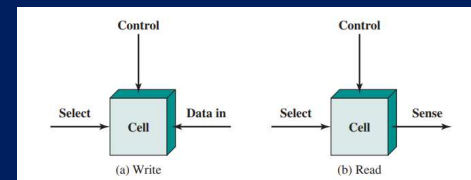
- **Storage capacity**: It generally refers to the number of storage units included in a memory.
- **Access time** : For **random access memory**, access time is the **time it takes to perform a read or write operation**. Whereas for **non-random access**, it is the **time needed to position read / write mechanism at desired location**.
- **Memory cycle time** : access time plus transient time (any additional time required before a second access can commence). Time may be required for the memory to “recover” before next access. Cycle time is **access + recovery**.

Technical Indicator of Memory Storages

- **Access rate**: The total number of bits of information exchanged between main memory and the outside (such as CPU) per unit time.
- **Reliability**: It is described by the mean time between failures; it is the average interval time between two failures.

Semiconductor Memory

Memory Cell – it is the basic element of a semiconductor memory



Two Types of Semiconductor Memory

- Random Access Memory
- Read-Only Memory

Random Access Memory (RAM)

- The contents of the memory unit can be freely taken out or stored as needed
- It will lose its memory contents when the power is turned off.
- It is mainly used to store programs for short-term use

Volatile Memory

DRAM (Dynamic random-access memory)
 FPM DRAM (Fast page mode DRAM)
 EDO DRAM (Extended data out DRAM)
 VRAM (Video random access memory)
 SDRAM (Synchronous dynamic random-access memory)
 DDR SDRAM (Double data rate SDRAM)
 DDR2 SDRAM, DDR3 SDRAM, DDR4 SDRAM
 RDRAM (Rambus DRAM)
 XDR DRAM (Extreme data rate DRAM)
 SGRAM (Synchronous graphics RAM)
 GDDR SDRAM (Graphics DDR SDRAM)
 GDDR2, GDDR3 SDRAM, GDDR4 SDRAM, GDDR5 SDRAM, GDDR6 SDRAM
 HBM (High Bandwidth Memory)
 PSRAM (Pseudostatic RAM)
SRAM (Static random-access memory)
CAM (Content-addressable memory)

Read Only Memory (ROM)

- This is designed to **hold permanent data**, and in normal operation is only read from, not written to.
- It is usually used to store system software which must be immediately accessible to the computer, such as the **BIOS program which starts the computer**, and the software for portable devices and embedded computers such as microcontrollers.

Read Only Memory (ROM)

ROM (Read-only memory)

MROM (Mask programmed ROM or Mask ROM)

PROM (Programmable read-only memory)

EPROM (Erasable programmable read-only memory)

EEPROM (Electrically erasable programmable read-only memory)

NVROM (Non-volatile random-access memory)

FRAM (Ferroelectric RAM)

Flash Memory

Flash Memory

combines the advantages of ROM and RAM, not only with the ability of electronic erasable programmable, but also does not loss data when power off and can quickly read data after power on.

Semiconductor Memory Types

Memory Type	Category	Erasure	Write Mechanism	Volatility
Random-access memory (RAM)	Read-write memory	Electrically, byte-level	Electrically	Volatile
Read-only memory (ROM)	Read-only memory	Not possible	Masks	Nonvolatile
Programmable ROM (PROM)				
Erasable PROM (EPROM)	Read-mostly memory	UV light, chip-level	Electrically	
Electrically Erasable PROM (EEPROM)		Electrically, byte-level		
Flash memory		Electrically, block-level		

Semiconductor Memory Errors

- Hard Failure
- Soft Errors

❖ Both hard and soft errors are clearly undesirable, and most modern main memory systems include logic for both detecting and correcting errors.

Hard Failure

Hard failure - is a **permanent physical defect** so that the memory cell or cells affected cannot reliably store data but become stuck at 0 or 1 or switch erratically between 0 and 1.

- ❖ Hard errors can be caused by harsh environmental abuse, manufacturing defects, and wear-out.

Soft Error

Soft error is a **random, nondestructive event** that alters the contents of one or more memory cells without damaging the memory.

- ❖ Soft errors can be caused by power supply problems or alpha particles. These particles result from radioactive decay and are distressingly common because radioactive nuclei are found in small quantities in nearly all materials.

Thank You
